

Business Analytics – SAS Base

Lecture 3: Data exploration

Aim of the lecture

- Elaborate what data exploration is and its importance in real life
- Walk you through the main data exploration functions and the related statistics concepts

Agenda

- **Data exploration intro**
- Data exploration functions
- Recap

Data exploration could be the main responsibility of an analyst

What are the applications of data exploration?

Data Quality Control

- Make sure that the data collected has good quality and can support the needs for analytics

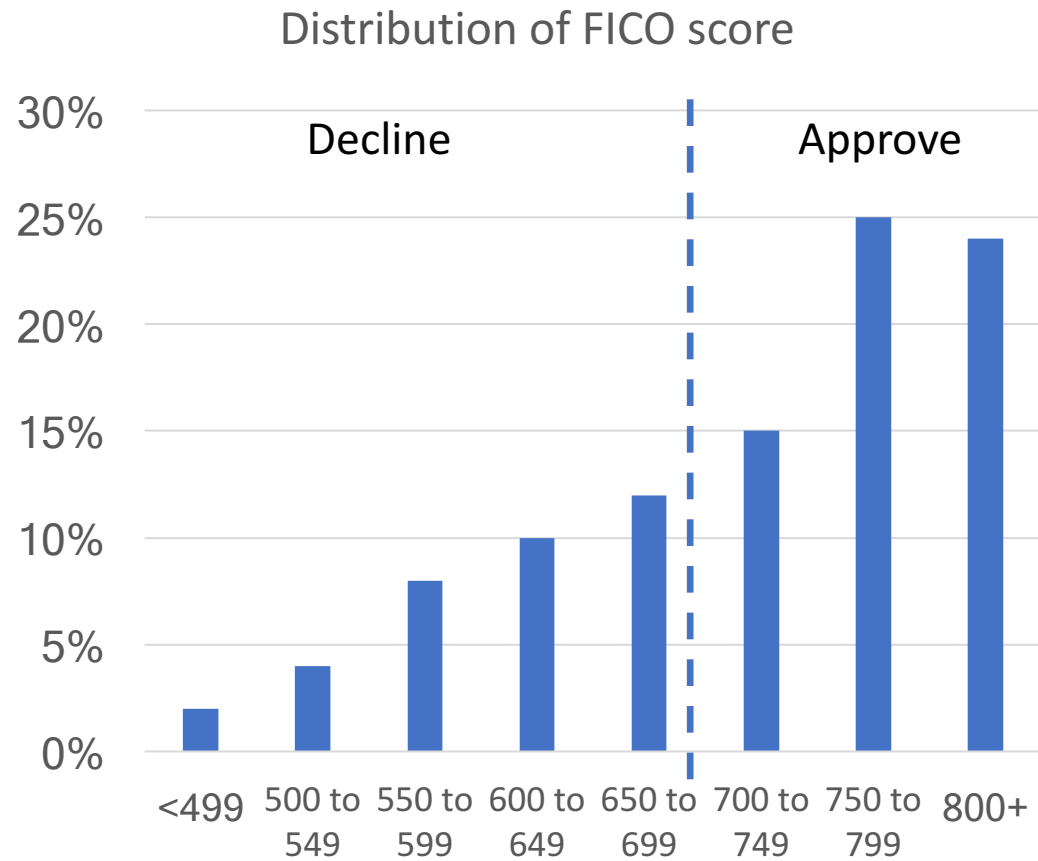
Preprocessing the data for predictive modeling

- During data exploration, some data cleaning and manipulation can be performed for the modeling work

Analytics projects

- Some business analytics projects can be completed with the data exploration functions in SAS directly
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Using data without proper exploration can be detrimental to a data driven strategy

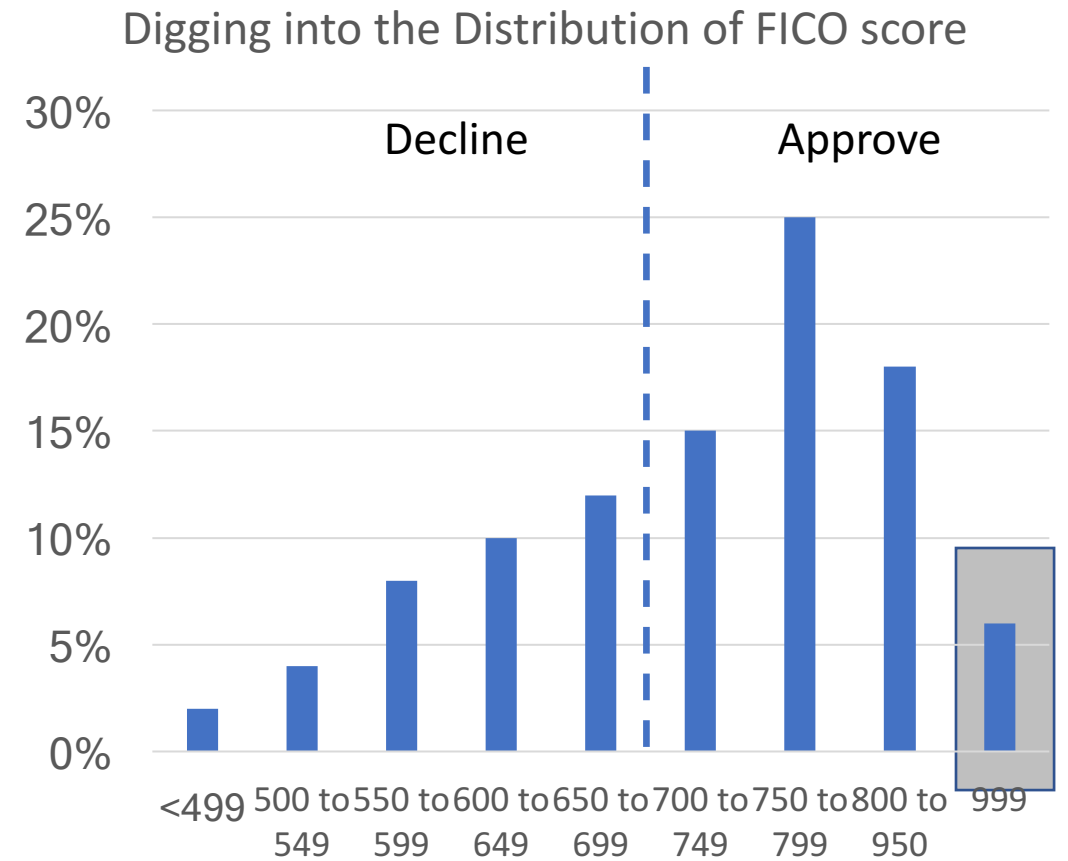
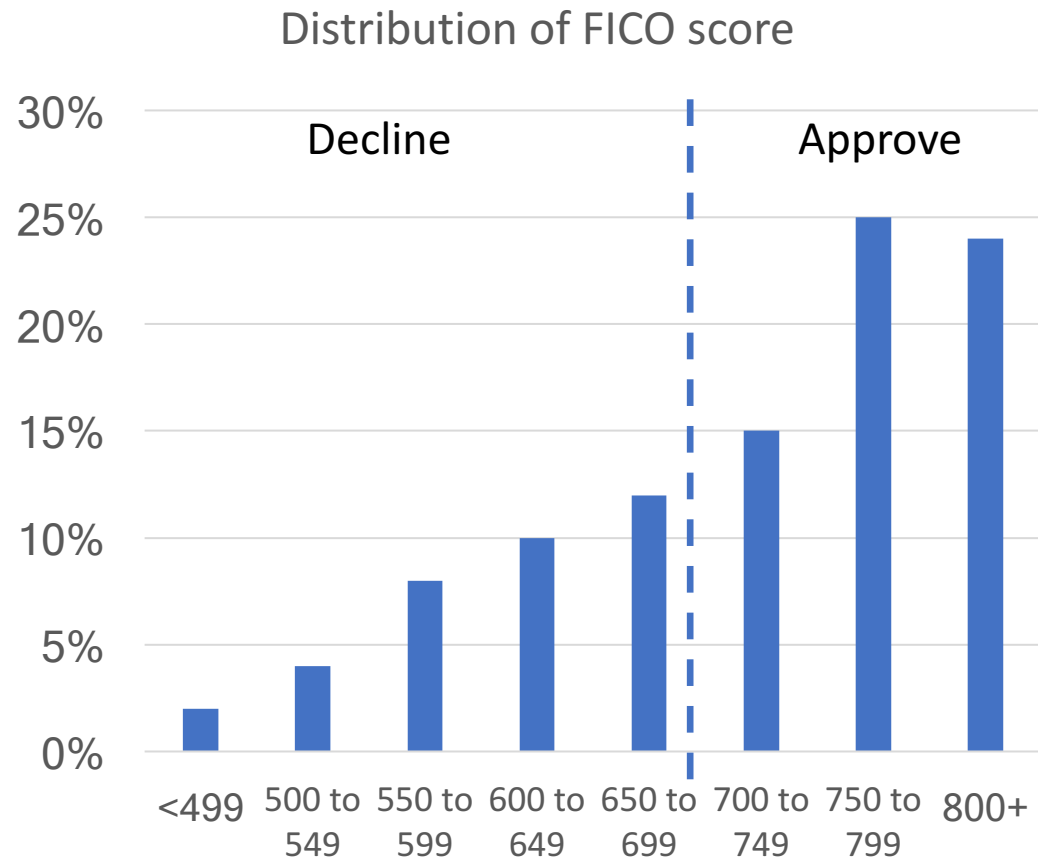


Looks solid!

But the strategy kept losing money...



Using data without proper exploration can be detrimental to a data driven strategy



Agenda

- Data exploration intro
- **Data exploration functions**
- Recap

There are four main data exploration functions in SAS

Function	Useful for...
Proc contents	Understanding the structure of the data
Proc freq	Distribution of categorical variables
Proc means	Summary statistics for numeric variables
Proc univariate	Custom percentile, histogram, etc.

This is a list to help you memorize, details will follow

PROC CONTENTS function returns the structure of the data

Method

```
PROC CONTENTS DATA = data_input  
OUT = result_output noprint;  
RUN;
```

My recommendation is to run this whenever you get a new dataset

PROC FREQ function shows the distribution of categorical variables

Method (one variable)

```
PROC FREQ DATA = data_input;  
TABLE variable /nocum nopercnt norow;  
OUT = result_output;  
RUN;
```

Method (multiple variables)

```
PROC FREQ DATA = data_input;  
TABLE variable1*variable2 /list nocum nopercnt norow;  
OUT = result_output;  
RUN;
```

PROC FREQ function shows the distribution of categorical variables

Method (one variable)

```
PROC FREQ DATA = data_input;  
TABLE variable /nocum nopercnt norow;  
OUT = result_output;  
RUN;
```

Method (multiple variables)

```
PROC FREQ DATA = data_input;  
TABLE variable1*variable2 /list nocum nopercnt norow;  
OUT = result_output;  
RUN;
```

PROC MEANS function shows the statistical properties of numerical variables

Statistical Option	Description
N	Number of observations
NMISS	Number of missing observations
MEAN	Arithmetic average
STD	Standard Deviation
MIN	Minimum
MAX	Maximum

Method

PROC MEANS

DATA = data_input;

WHERE logic for data selection;

CLASS categorical_variable;

VAR variable1 variable2 ...;

OUT = result_output;

RUN;

PROC UNIVARIATE function shows more detailed statistical properties

Statistical Option	Description
N	Number of observations
NMISS	Number of missing observations
MEAN	Arithmetic average
STD	Standard Deviation
MIN	Minimum
MAX	Maximum
SUM	Sum of observations
MEDIAN	50th percentile
Pn/Qn	nth percentile/quartile

Method

```
PROC UNIVARIATE  
DATA = data_input;  
WHERE logic for data selection;  
CLASS categorical_variable;  
VAR variable1 variable2 ...;  
OUT = result_output;  
HISTOGRAM;  
RUN;
```

Recap: Now you know how to use SAS for data exploration

- Data exploration is very important in both analytics and modeling work
- The four important data exploration functions are proc contents, proc freq, proc means, and proc univariate
- Homework for today is in fact an analytics project, enjoy! 😊

*Thank
you*

